Design and Implementation of Artix-7 FPGA-Based Vending Machine with Debouncing

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Introduction:

Vending machines are automated systems that dispense products based on user input and payment. In this project, a vending machine is implemented using **FPGA (Field Programmable Gate Array)** and **Verilog HDL**.
The design focuses on:

- Accurate money handling
- Reliable product dispensing
- Stock management
- Eliminating button noise using debouncing

This system is implemented on the **Basys 3 FPGA board**, where switches and push buttons act as inputs and LEDs indicate outputs.

Objective:

The main objectives of this project are:

- To design a vending machine using digital logic
- To implement the design using Verilog HDL
- To simulate the system using a testbench
- To deploy the design on FPGA hardware
- To eliminate noisy input signals using a debounce circuit

System Description:

The system consists of three main modules:

1. Vending Machine Module
2. Debounce Module
3. Testbench for Simulation

Product and Pricing Details:

The vending machine supports four products:

Product	Selection Code	Price

Gum	1	25 cents
Chocolate	10	75 cents
Chips	100	$1.5
Drink	1000	$2

Each product tray initially contains: 15 items

Working Principle:

The vending machine operates as a sequential digital system based on user inputs.

Step-by-step operation:

Money Insertion

- quarter input → adds 25 cents
- dollar input → adds 100 cents or 1 dollar

Money Storage

- Stored in a register: Money

Product Selection

- User selects product using: select[3:0]

Purchase Operation

- When buy button is pressed:
  - System checks:
    - Is money sufficient?
    - Is product available?

Product Dispensing

- If valid:
  - Product signal becomes HIGH
  - Money is deducted
  - Stock decreases

Out-of-Stock Handling

- If stock becomes zero: out_of_stock[x] = 1

Restocking

- Using load input, stock resets to 15

Edge Detection Logic:

To avoid multiple increments from a single press, edge detection is used:

quarter_prev, dollar_prev, buy_prev
This ensures:

- Action occurs only when signal changes: 0 → 1

Debouncing Concept:

Problem

Push buttons generate noisy signals due to mechanical bouncing:
0 → 1 → 0 → 1 → unstable
This can cause:

- Multiple purchases
- Incorrect money updates

Solution: Debounce Module

The debounce module:

- Filters noise using a counter
- Produces a clean signal

```
if (count == 16'hFFFF)
   btn_clean <= btn_sync;
```

Usage in this project:

Applied to:

  - Buy button
  - Reset button

System Architecture:

Switches / Buttons → Debounce → Vending Machine → LEDs

Verilog Implementation:

(a) Vending Machine Module

Features:

- Money calculation
- Product dispensing
- Stock management
- Restocking

(b) Debounce Module

- Synchronizes input
- Uses counter-based filtering
- Outputs stable signal

(c) Testbench

Used to verify system functionality before hardware implementation.
Example:

```
#10 dollar = 1;
#10 dollar = 0;
```

Simulation Results:

Simulation was performed using Vivado.
Observed Behavior:

- Clock signal toggles continuously
- Money increases correctly on input
- Product output activates when conditions are met
- Money decreases after purchase
- Stock reduces after each transaction
- Out-of-stock signal activates when stock is empty

Observation:

- Product signal is a short pulse
- Requires waveform zoom to observe

Hardware Interaction:

Input Signal	FPGA Component
clk	Onboard 100 MHz Clock
btnU	Push Button (Reset)
btnC	Push Button (Buy)
btnR	Push Button (₹25 Input)
btnL	Push Button (₹100 Input)
sw[3:0]	Slide Switches (Product Selection)
sw[7:4]	Slide Switches (Restocking)

OUTPUT MAPPING:

Output Signal	FPGA Component
led[3:0]	LEDs (Product Dispensed)
led[7:4]	LEDs (Out-of-Stock Indicator)

Constraints (XDC File):

Constraints map logical signals to FPGA pins.
Example:
set_property PACKAGE_PIN V17 [get_ports quarter]
Purpose:

- Connect switches and buttons to inputs

- Connect LEDs to outputs

Advantages:

- Reliable input handling using debounce
- Simple and modular design
- Easy to expand for more products
- Suitable for real-time hardware implementation

Conclusion:

The vending machine was successfully implemented using FPGA and Verilog HDL. The system correctly handles money input, product selection, stock management, and dispensing operations. The debounce module ensures reliable input handling, making the system robust and suitable for real-world applications.